LOGISTIC REGRESSION PROJECT REPORT

Project: Diabetes Prediction using Kaggle Data
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Dataset: Diabetes predication system with KNN algorithm
Link: https://www.kaggle.com/datasets/abdallamahgoub/diabetes
File: diabetes.csv

Model Used:

For this project, I used **Logistic Regression** to predict whether a person is diabetic or not. Before training the model, I followed a few preprocessing steps:

- **Median Imputation** – Missing values were filled using the median value of the respective feature.
- **StandardScaler** – The numerical values were scaled so that all features were on a similar scale.
- **Balanced Logistic Regression** – The model was balanced to handle the difference between diabetic and non-diabetic cases.

Objective:

The main objective of this project is to develop a Logistic Regression model to predict whether a person has diabetes or not based on different health-related factors. The target variable used in this project is Predict Outcome, where:

- 0 represents No Diabetes
- 1 represents Diabetes

This project also focuses on understanding the dataset, handling duplicate, invalid and missing values, preparing the data for machine learning, and evaluating the performance of the model. In addition, the project aims to identify which factors, such as glucose level, BMI, age, and insulin level, have a greater impact on diabetes prediction.

Dataset and Workflow:

For this project, I have used the Diabetes Dataset from Kaggle. The dataset contains different health-related information of people, which is used to predict whether a person has diabetes or not. I have used the KNN (K-Nearest Neighbors) algorithm for making the prediction.

Input Features:

The dataset contains the following features:

- **Pregnancies** – Number of times a person has been pregnant.
- **Glucose** – Glucose level in the blood.
- **BloodPressure** – Blood pressure level.
- **SkinThickness** – Thickness of the skin fold.
- **Insulin** – Insulin level in the blood.
- **BMI** – Body Mass Index.

- **Pedigree** – Shows the effect of family history of diabetes.
- **Age** – Age of the person.

The **Outcome** column is used for prediction. A value of `0` means the person does not have diabetes, while `1` means the person has diabetes.

Workflow:-

I followed the following steps to complete the project:

Load the Dataset → Understand the Data → Check Missing Values and Duplicates → Perform Data Analysis → Analyze the Distribution of Features → Check Correlations → Detect and Treat Outliers → Separate Input and Target Variables → Split the Data into Training and Testing Sets → Balance the Training Data using SMOTE → Scale the Features → Train the Logistic Regression Model → Make Predictions → Evaluate the Model using Accuracy, Precision, Recall, F1-Score and Confusion Matrix.

First, I loaded the dataset and checked its structure, data types, missing values, and duplicate records. Then, I performed univariate, bivariate, and multivariate analysis to understand the data better.

After that, I detected outliers using the IQR (Interquartile Range) method and capped the extreme values. The data was then divided into training and testing sets in an 80:20 ratio.

Since the target classes were not equally distributed, I used SMOTE to balance the training data. Finally, I used StandardScaler to scale the features and trained a Logistic Regression model. The model was then evaluated using different performance measures such as accuracy, precision, recall, F1-score, and confusion matrix.

Evaluation Metrics:-

After training the model, I evaluated its performance using different evaluation metrics. These metrics help to understand how well the model is predicting whether a person has diabetes or not.

Metric	Result	Meaning
Accuracy	**68.83%**	About 68.83% of the predictions were correct.
Precision	**54.79%**	About 54.79% of the people predicted as diabetic were actually diabetic.
Recall	**72.73%**	The model correctly detected about 72.73% of the actual diabetic cases.
F1-score	**62.50%**	Shows the balance between precision and recall.

Classification Report:

Class	Precision	Recall	F1-score	Support
0 – No Diabetes	0.81	0.67	0.73	99
1 – Diabetes	0.55	0.73	0.62	55

Accuracy			**0.69**	**154**
Macro Average	0.68	0.70	0.68	154
Weighted Average	0.72	0.69	0.69	154

Interpretation

The model achieved an overall accuracy of 68.83%, which means that around 69 out of 100 predictions were correct. The recall for the diabetic class was 72.73%, so the model was able to detect most of the people who actually had diabetes.

The precision was 54.79%, which means that some people predicted as diabetic were actually not diabetic. The F1-score of 62.50% shows a balance between the model's precision and recall.

Overall, the model was able to make fairly good predictions on the test data. However, its performance can still be improved by using better preprocessing, feature selection, or trying other machine learning algorithms.

Confusion Matrix:-

The confusion matrix shows the number of correct and incorrect predictions made by the model.

	Predicted 0	**Predicted 1**
Actual 0	TN = **66**	FP = **33**
Actual 1	FN = **15**	TP = **40**

Interpretation

- **True Negative (TN = 66):** 66 people who did not have diabetes were correctly predicted as non-diabetic.
- **False Positive (FP = 33):** 33 people who did not have diabetes were incorrectly predicted as diabetic.
- **False Negative (FN = 15):** 15 people who had diabetes were incorrectly predicted as non-diabetic.
- **True Positive (TP = 40):** 40 people who had diabetes were correctly predicted as diabetic.

Key Findings

1. Glucose level was found to be one of the important factors for predicting diabetes.
2. The dataset contains 768 records, and the number of non-diabetic people is higher than the number of diabetic people.
3. Around 34.90% of the people in the dataset are diabetic, while 65.10% are non-diabetic.
4. Some outliers were found in features such as Pregnancies, Glucose, BloodPressure, Insulin, BMI, Skin Thickness, Age and Pedigree. These outliers were treated using the IQR method.
5. SMOTE was used to balance the diabetic and non-diabetic classes in the training data.

6. The Logistic Regression model achieved an accuracy of 68.83% on the test data.
7. The model achieved a 72.73% recall for diabetic cases, which means it was able to identify most of the actual diabetic cases.
8. Overall, the model was able to make useful diabetes predictions, but its performance can be further improved by trying different algorithms and improving the data preprocessing.

Limitations

1. The model achieved 68.83% accuracy, which means some of the predictions were still incorrect.
2. The precision for diabetic cases was 54.79%, so some people were predicted to have diabetes even though they were actually non-diabetic.
3. The dataset has only 768 records, which is relatively small and may not represent different types of people and populations.
4. The model uses only 8 health-related features. Other important factors that may affect diabetes are not included in the dataset.
5. SMOTE was used to balance the training data, but the additional samples created by SMOTE may not be exactly the same as real patient data.
6. Since the project uses Logistic Regression, it may not be able to capture some complex relationships between the different health factors.
7. This system is mainly designed for prediction and learning purposes and should not be used as a replacement for proper medical testing or professional diagnosis.

Conclusion

In this project, I developed a Diabetes Prediction System using Logistic Regression. The main aim was to predict whether a person is likely to have diabetes based on different health-related factors such as Glucose, BloodPressure, BMI, Age, Insulin, and other features.

During the project, I performed data analysis, checked missing values and duplicates, identified and treated outliers, and used SMOTE to balance the training data. I also applied StandardScaler before training the model. The Logistic Regression model achieved an accuracy of 68.83% and a recall of 72.73% for diabetic cases. These results show that the model was able to identify a good number of actual diabetic cases from the test data.

Overall, this project helped me understand the complete machine learning process, from data preprocessing and analysis to model training and evaluation. The system can be further improved by using a larger dataset, additional health-related features, and other machine learning algorithms to achieve better results.